

What a domain prior can and cannot replace

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TGDC case study, research track

Abstract

Whether hand-written rules can stand in for a client’s operational records when generating enterprise data is tested here on 311 tickets from four cities. Blind rules clear a structural null in San Francisco (p 0.0015) and not in Austin (p 0.110), and a conditional fitted on the real data exceeds them in both. Two cheaper references carry most of the signal: category volume alone scores 0.672 against 0.676 for the author-written rules, and a city’s published service target recovers 79% of the achievable margin with no records at all. An agent benchmark built on generated tickets therefore tests a much weaker problem than it appears to.

Introduction

Generated enterprise data is built from a description of a business rather than from anyone’s real records: an org chart, a list of the work, and hand-written rules for how that work behaves. The appeal is that a client unable to share data can still be served, and the open question is what that substitution costs. Distributional fidelity is readily measured but is already established as a poor guide to downstream usefulness, so the operative question is which properties of real records the rules can supply and which must come from the client.

This study draws on public 311 service request feeds from four cities. Each ticket is treated as a unit of work and the outcome of interest is time to closure. A service catalogue lists job types and the departments that own them with no timings attached, so it stands in for the description of a business, while the recorded durations stand in for the client records a generator does without.

Hypothesis

The hypothesis under test is that rules reasoned from the nature of the work, written without sight of any outcome, predict time to closure at an AUC between 0.60 and 0.72, and that any value below 0.55 would indicate that the reasoning contributes nothing.

Related work

That fidelity and utility can come apart is established. Karr et al. (2006) and Snoke et al. (2018) set out utility measures that rank generators differently [1, 2]; Hansen et al. (2023) report high fidelity alongside poor downstream performance [3]; van Breugel et al. (2023) show that models trained on synthetic data without care do not carry back to real data [4]; and Du and Li (2025) argue that the standard metrics may not measure the right thing [5]. That divergence is taken as given here.

The nearer literature treats language models as priors over tabular problems, of which this study is a small instance. Knauer et al. (2025) induce a decision tree from the target and the column names alone, with no rows and no labels [6], as the model-written rules do here; Hegsel-

mann et al. (2023) show with TabLLM that column names carry zero-shot signal before any label [7]; and Capstick et al. (2025) score an elicited prior by marginal likelihood in AutoElicit, choosing between priors by Bayes factor [8], which needs no test split and is the procedure this study would adopt next time. Previous work is confined to classification where the target here is duration, and the first three need descriptive feature names, a precondition Knauer et al. state. A 311 taxonomy is unusually kind on that point; an enterprise schema of coded columns would not be.

Method

The records are written by the people who do the work. Each feed reports the work type, the owning department, the intake channel and the timestamps. The full weeks of 4 May and 8 June 2026 were taken from each city (Table 1).

Tickets still open at the time of extraction were retained and coded as slow, since all lie months beyond any threshold; Chicago closes most of its feed within a second of creation, and those rows were dropped as records rather than work.

The task is to predict, at the moment a ticket opens, whether closure will take longer than the median. Let A denote the AUC attained by an arm and C the ceiling AUC available from the same columns; recovered skill, reported alongside AUC, is then $S = (A - 0.5) / (C - 0.5)$. The null tested is that the rules carry no information about which work runs slow, so that any score they reach is available from carving up the same taxonomy at random.

A ceiling is required for the denominator. Since a prior reads two columns, the service category and the owning department, the conditional rate was fitted from those same two to establish what they support. It is a taxonomy ceiling, not a general limit on skill, and it separates a weak rule from weak features. The estimator was settled on after the first Austin arm had been scored against a different one, so the AUCs reported here are pre-registered and the denominator is not; every percentage sits beside its raw AUC.

Two kinds of interval are reported. Resampling tickets treats 17,007 San Francisco rows as independent when the rules were written per category, so categories were resampled as well [9]. That second unit carries two warnings. A score is constant within a category only where the category has one owning department, which holds throughout Austin but fails for 22 of San Francisco’s 37 categories, 99.6% of its rows. Clustered resampling is also asymptotic in the number of clusters and assumes rough balance, where one San Francisco category holds 34% of the rows and the effective count is 5.3 against 37 categories, Austin 22.3 against 117. Almost nothing could clear an interval that wide, so the category intervals are descriptive rather than tests; where a result below is called not established, the meaning is that this design cannot

city	train	test	median resolution	categories
New York	72,206	78,296	5.4h	164
Chicago	17,291	38,671	118.0h	92
Austin	5,968	6,029	23.8h	117
San Francisco	17,723	17,007	15.1h	37

Table 1: the four feeds, two complete weeks from each.

establish it [10].

Each catalogue was pulled with the query limited to category names and row counts, so no duration, rate or outcome reached the author. The rules were written from that alone, committed with a prediction, and only then evaluated. One San Francisco file was amended 27 seconds after commit, to match on word boundaries rather than substrings, before that city’s data existed on disk.

The principal risk is that the result describes one author’s reasoning, so Austin and San Francisco each carry two rule sets under the same blind condition, one by the author and one by a language model from the same catalogue. New York serves as the contrast, written under sight of the real rates, and Chicago as the transfer target for those rules.

Results

Blind prior recovery

The answer is not categorical (Table 2). The best blind rules recover 72% of the taxonomy ceiling in San Francisco and the weakest recover nothing in Austin; a single-city design would have supported a confident conclusion in either direction, the city deciding which.

The two intervals disagree. By ticket, three of the four arms exceed chance; by category, only the two San Francisco arms do, and Austin’s model-written set spans [0.442, 0.770], which is not distinguishable from guessing. The 72% spans roughly 7% to 93% on that unit, and every claim below should be read at that width.

Gain over structure

A rule set may score well because it encodes reasoning, or because almost any sensible cut of a lopsided taxonomy would; two controls separate the cases.

Shuffling the verdicts across categories preserves the taxonomy, the row counts and the set of scores, and breaks only which category received which score, so it draws from the null directly and the share of shuffles reaching the observed AUC is a permutation p-value (Table 3). The null assigns one verdict per category, so the arm is scored on that unit too, which is why the table carries a second AUC for it. The rules reach 0.741 in San Francisco with 2 of 2,000 shuffles matching, p of 0.0015, and 0.617 in Austin with 220 matching, p of 0.110. The null is therefore rejected in San Francisco and not in Austin: the reasoning does real work in one city and is not distinguishable from a lucky cut of the taxonomy in the other.

Frequency is the harder baseline, because the pull supplied row counts and those are themselves real operational data. Scored rarer-is-slower it reaches 0.672 against 0.676 for the author-written San Francisco rules, which therefore add almost nothing beyond volume; the model-written rules clear it by 0.105. The direction of that baseline is one bit the catalogue did not supply, so the comparison is generous to it if anything.

The author effect

The model-written rules exceed the author-written rules by 0.116 in Austin and 0.101 in San Francisco, the same direction at a similar magnitude in both cities. By ticket, both differences exclude zero, at [0.104, 0.131] and [0.095, 0.107]; by category, neither does, at [-0.036, 0.288] and [-0.021, 0.174]. Two cities are not sufficient, and an earlier draft reported these differences with no interval at all.

Pre-registered predictions

arm	predicted	observed	in band
Austin, author	0.60 to 0.72	0.494	no
Austin, model	0.52 to 0.62	0.611	yes
SF, author	0.55 to 0.68	0.676	yes
SF, model	none	0.777	-

The first band was set before the project had produced any result, and it was wrong. The next two were set after Austin had been observed, so the expected range was already known, a weaker achievement and counted as one. The fourth arm produces the headline 72% and carries no prediction at all. The only prediction made in genuine ignorance is the one that failed.

One department

Austin Resource Recovery (Table 4) accounts for 34% of the city’s volume and 70% of its work runs slow, yet both rule sets classify it as fast, 0.37 for the model and 0.05 for the author. With the department removed, the model reaches 70% against San Francisco’s 72% and the author 38% against 46%, so the gap closes for both. A missed collection remains open until the next scheduled route, a week away: the work is routine while the ticket is not, and nothing in the phrase ARR - Compost conveys this.

Applied to San Francisco, the same operation cuts the other way, which is the more important half of the table. Its two largest categories are 57% of test volume, both fast and both called fast. With them removed the ceiling barely moves, 0.886 to 0.851, while the model falls from 72% to 27% and the author from 46% to 20%. A two-line rule calling those two fast and the rest slow scores 0.699 by itself, so the 72% rests on a few high-volume decisions rather than on 37 categories of reasoning.

The null was tested on these subsets too, and it cuts both ways. Austin without ARR rejects at p 0.0025, against 0.110 for the whole city; San Francisco without its top two stops rejecting, at p 0.1374 against 0.0015. The one significant result does not survive losing the two categories that carry it.

ARR is a department prefix rather than a category called waste, so removing it also takes out fast work it owns, such as ARR - Dead Animal Collection at 0.0% slow. Graffiti Public accounts for 8.2% of San Francisco volume and is 85.1% slow, yet is priced at 0.37 by the model; the failure ARR shows in Austin is therefore present there too, and no subset here removes it.

city	rules	AUC	by ticket	by category	ceiling	recovered
Austin	author	0.494	[0.481, 0.508]	[0.331, 0.649]	0.883	-1%
Austin	model	0.611	[0.598, 0.626]	[0.442, 0.770]	0.883	29%
San Francisco	author	0.676	[0.670, 0.684]	[0.515, 0.815]	0.886	46%
San Francisco	model	0.777	[0.770, 0.784]	[0.529, 0.859]	0.886	72%

Table 2: what blind rules recover, by city and by author. Ticket intervals are 1,000 resamples, category intervals 2,000.

city	arm	AUC	[95% CI]	recovered	p vs H0
Austin	blind rules, per ticket	0.611		29%	
Austin	blind rules, per category	0.617			0.110
Austin	category frequency alone	0.516	[0.351, 0.726]	4%	
Austin	same rules, verdicts shuffled	0.501	[0.322, 0.674]	0%	
SF	blind rules, per ticket	0.777		72%	
SF	blind rules, per category	0.741			0.0015
SF	category frequency alone	0.672	[0.441, 0.785]	45%	
SF	same rules, verdicts shuffled	0.499	[0.312, 0.690]	0%	

Table 3: blind rules against two baselines built from structure alone, with category intervals and the permutation test of the null.

The real-data comparator

Given the same two columns, a conditional fitted on the real data exceeds the blind rules by 0.108 in San Francisco and 0.272 in Austin, so real records win on this task in both cities. Against a conditional using the category column alone the rules appear level, 0.777 to 0.771, but that comparator lacks the department column the rules read, which is worth 0.115 in San Francisco, and the 0.0067 margin has a paired interval of [-0.201, +0.108].

The finding

The finding is stated as an ordering rather than a level, because the level moves with the categories in the subset and the interval on it is wide. Blind rules exceed chance in San Francisco and not in Austin; the model-written rules exceed the author-written rules in both cities; and removing one administratively clocked department closes most of the gap between the cities for both authors.

The explanation offered for that ordering is that a blind prior holds where duration follows from the job and fails where the clock is an administrative cycle. The partition was derived from the categories the author-written rules got wrong, so it was built to fit them. On the subset it names, however, the null is rejected at $p = 0.0025$, which it was not fitted to produce. A fifth city, with those categories named from the taxonomy before any measurement, is the test that could still fail.

Value of the real rates

Rules written with a city’s rates in hand reach 81% there, above the 72% a blind author obtained in San Francisco but not by much.

The last row was not expected. New York publishes the target it holds itself to, in days, per complaint type. Ranking tickets by that alone reaches 0.830, or 79%, matching rules written under sight of the real outcomes and needing no reasoning, no language model and no client records. Where a client publishes what they intend to take, that document is worth about as much as the exercise this paper measures, and is the first thing to ask for.

It bears on contamination too, since a language model may be recalling published material rather than reasoning. No equivalent dataset was found for Austin or San Francisco, which narrows that concern without settling it.

Carried to Chicago at Chicago’s own median the rules score 0.496, which an earlier draft called actively misleading. That does not hold (Table 5): the medians are 5.4h and 118.0h, so the carried row poses a question the rules were not written for. At New York’s own threshold they reach 0.581, or 37%. The remaining shortfall is mechanical, since they fire on 17.1% of Chicago rows, being New York acronyms.

Distribution fidelity

The first recorded hypothesis, fixed in `HYPOTHESIS.md` before any run, was that generated records matching real data to low-order fidelity would still hold combinations the real process could never produce, and that accuracy would fall as those rose. It was tested by fitting generators on real New York records, sampling, training on the sample and testing on real records [11]. It failed, for a structural reason. The pairwise generator draws department conditional on category, so it cannot emit a pair it never saw and its 0.0% impossible rate is an identity; the learner is additive over one-hot columns with no interaction terms, so a wrong combination could not have hurt it either. Both were settled by the design before any data was touched. The work moved to the narrower question this paper answers, which is why the stated hypothesis and the pre-registration are not the same sentence.

The ladder still shows the one thing it is here for. Marginals alone lose almost everything, 0.511 against a 0.918 ceiling, while adding pairwise dependence recovers all of it at 0.917. A generator clearing the column shapes and column pair trends a quality report scores [12] is indistinguishable from real records here, so a distribution-level check cannot see the effect this paper measures. The generators used here are conditional probability tables with no privacy noise, so they are not PrivBayes or MST, but the order of structure they keep is the order those methods fit [13, 14].

Limits

Four cities, one domain, one task, and only two of the cities blind. New York is a contamination control and Chicago its transfer target, so neither replicates anything, and what does replicate is an author effect whose category

subset	rules	n	prior	ceiling	recovered
Austin, all	author	6,029	0.494	0.883	−1%
Austin, all	model	6,029	0.611	0.883	29%
Austin, minus ARR	author	4,008	0.644	0.881	38%
Austin, minus ARR	model	4,008	0.766	0.881	70%
SF, all	author	17,007	0.676	0.886	46%
SF, all	model	17,007	0.777	0.886	72%
SF, minus top 2	author	7,255	0.571	0.851	20%
SF, minus top 2	model	7,255	0.595	0.851	27%

Table 4: dropping the highest-volume categories in each city, for both sets of rules. The same operation run both ways.

rules	tested on	threshold	AUC	ceiling	recovered
New York	New York	5.4h	0.840	0.918	81%
NY rules carried	Chicago	118.0h	0.496	0.717	−2%
NY rules carried	Chicago	5.4h	0.581	0.717	37%
published target	New York	5.4h	0.830	0.918	79%

Table 5: rules written with the rates in hand, those rules carried to another city, and the city’s own published service target.

intervals cross zero. A 311 feed offers three usable columns, no documents that must agree and no prices, so it is a thin stand-in for enterprise data.

The second author is a language model, which is the largest weakness. These feeds are widely mirrored, so blind here means the session was shown no durations, not that the weights hold none: language models memorise popular tabular datasets and score better on ones they have seen [15], the exact failure this arm admits, and it produces every headline number. Neither the prompt nor the model version was recorded, so it cannot be rerun.

The Austin arm also has an ordering problem. The commit that evaluated the author-written rules names compost at 96%, traffic signal maintenance at 8% and vehicle abatement at 94%, all three called the wrong way. The model wrote its Austin rules twelve hours later, using the author’s own phrasing as keyword strings, and got two of the three right. Compost it still called fast, which argues against wholesale transfer, but the prompt was not recorded and this cannot be settled. The blind condition covers the catalogue pull, not what had already been written down.

The San Francisco rules were written in knowledge of what Austin had shown. New York carries known artefacts too, including a department that closes most tickets at exactly midnight.

The general question

The first step is to make the rules generate. The arms here score rule sets against a conditional fitted on real data, which measures how much of a conditional a person can guess, not what generated data costs. Sampling rows from the taxonomy and volume counts, labelling them with the prior, then training on those and testing on real records would answer it directly. The second is the fifth-city test named above, which is the only way the administrative-clock partition can still fail.

This bears on the evaluation of agents. Where a taxonomy and a published target reach most of the achievable skill, an agent scored on tickets generated from that taxonomy is scored on the easiest part of the problem to reconstruct, and the categories where a blind prior fails are invisible in the taxonomy it was built from. A high

score there is evidence about the generator, not the agent.

A prior can be evaluated as a prior only once per dataset, before any outcome has been seen, and the prediction must come first. Of the predictions recorded here, the only one made in genuine ignorance is the one that proved wrong.

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